MEMORANDUM

TO: ebike conversion Contact list

FROM: Hassan, Ibrahim Khaleel

DATE: June-09-2025

RE: Report 1: Battery Pack Calculations

EXECUTIVE SUMMARY

The purpose of this report is to determine the appropriate battery pack configuration based on the load that will be attached to the battery pack. The desired battery pack specifications must satisfy the various requirements of the system and must not exceed the limits imposed by the electric e-bike motor, Battery Management System (BMS), and other relevant electrical components. The BMS is a device within the battery pack that is responsible for regulating the battery pack in order to prevent any overload or failure of the pack, which may result in fire. The BMS's main functions include overcurrent protection, undervoltage protection, overvoltage protection during charging, cell balancing, and temperature monitoring in some models [1]. The load in this application is a 350W e-bike motor with a voltage rating of 24V. The lithium-ion Samsung 50E 21700 rechargeable cell is the desired cell to build this battery pack. The specifications of this cell are shown below. As illustrated in Table 1, the Samsung 50E 21700 battery has a capacity of 5000mAh, nominal voltage of 3.6V, and a maximum continuous discharge rate (CDR) of 9.8A. Since the load is rated at 350W, the current needed by the motor was calculated using Equation 1,

$$P = V \times I \tag{1}$$

where, P is power rating of the load (motor), V is its voltage rating, and I is the current needed by the motor. I was then calculated to be 20A. Therefore, the battery pack needs to supply a total current of 20A to the motor at most. It is important to note that when the pack is at maximum

Specification	**Details**
Brand	Samsung
Model	50E
Battery Type	21700
Capacity	5000mAh
Nominal Voltage	3.6V
Full Charge Voltage	4.2V
Discharge Cutoff Voltage	2.5V
Continuous Discharge Rate (CDR)	9.8A
Rechargeable	Yes
Protected	No (Unprotected cell – use with proper safety measures)
Cycle Life	~300+ charge cycles
Diameter	21.4mm
Length	70.0mm
Weight	~70g

Table 1: Specifications of the Samsung 50E 21700 Battery Cell

voltage, i.e. 29.4V, less current is required by the motor (about 12A). Likewise, at a nominal voltage of 25.2V, a current of approximately 14A will be needed 3. This can be explained from the relationship between P, V, and I in Equation 1. The power needed by the motor does not change, so voltage and current vary at each point in time to ensure that the power supplied remains constant. Subsequently, using a total maximum current of 20A, the least number of cells to be connected in parallel was determined. A summary of the results in Table 2 shows 3 cells in parallel were needed to supply sufficient current, hence a 3P arrangement. Thus, the chosen pack design is a 7S3P battery configuration, consisting of 21 cells in total as shown below 1. In addition, the max BMS rating required was calculated to be 23A. The total amount of energy for the entire pack was

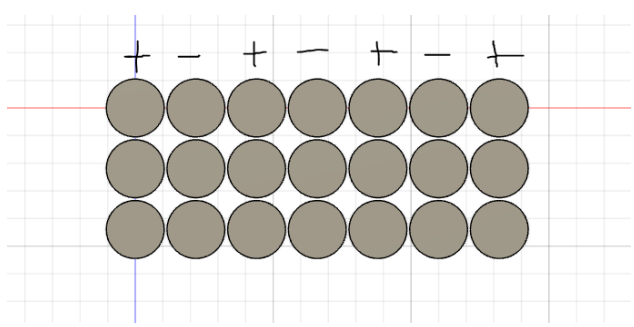

Figure 1: 7S3P Battery Pack Configuration

computed using the following equations.

$$E = P \times t \tag{2}$$

where, E is energy, P is power and t is time. Substituting Equation 1 into 2 gives,

$$E = V \times I[A] \times t[h] \tag{3}$$

It is important to note that battery capacity is defined as the amount of electrical charge stored in a battery [2]. Battery capacity is measured in Amp-hours (Ah). Its equation is given by,

$$Capacity[Ah] = I[A] \times t[h] \tag{4}$$

Using the equations above, the total battery pack energy was then calculated to be 360Wh. Table 2 below shows a summary of the results for the desired battery pack specifications. Based on

Parameter	**Value**
Pack configuration	7S3P
BMS rating	23A
Total Pack energy	360Wh
Nominal pack voltage	24V
Pack capacity	15Ah

Table 2: Summary of Results for the Battery Pack Configuration

these calculations, it is crucial to try and identify where potential errors may have occurred. For instance, no system component works optimally at 100% without inefficiencies or power loss. In electrical and electro-mechanical systems, power losses in the form of heat due to resistance in wires and friction in the case of the motor occur. As a result, when performing computations, it may be useful to increase the margin of error as done in the BMS calculation in order to account for inefficiencies. For example, rather than using 350W as load rating, 385W could have been used as some of the power drawn will be wasted as heat. However, since these calculations are performed using variables, it makes it easier to change the values when needed.

RESULTS

Based on motor specifications, the voltage required by the motor is 24V, which was then utilized to calculate the current needed for the motor. According to Table 1, nominal voltage of each cell is 3.6V and cutoff voltage is 2.5V. Recall that connecting cells in series increases voltage while putting them together in parallel increases current and current-carrying capacity. In order to achieve 24V, 7 cells were connected in series, giving a 7S configuration. As Table shows us below, adding 7 cells in series actually gives a summation of 25.2V. However, this is the closest possible number to 24V, thus it is referred to as a 24V pack.

$$\text{Max pack voltage} = 4.2 \times 7 = 29.4\,\text{V}$$
$$\text{Nominal pack voltage} = 3.6 \times 7 = 25.2\,\text{V}\ (\sim 24\,\text{V})$$
$$\text{Lowest pack voltage} = 2.5 \times 7 = 17.5\,\text{V}$$

Figure 2: Calculated maximum, nominal, and lowest voltages for a 7-series cell pack.

The current required for the ebike motor to operate at maximum, nominal and lowest voltage is calculated using Equation 1. The results are as follows.

Voltage level	**Current required**
29.4V (max)	11.9A
25.2V (nominal)	13.9
17.5V (min)	20A

Table 3: Current Requirements at Different Voltage Levels

In order to ensure the battery pack delivers the sufficient amount of current to the ebike motor at each point in time, a battery pack configuration that can supply up to 20A continuous is required. However, it is important to understand that a current as high as this is only needed when the battery voltage has dropped to its lowest value i.e. 17.5V, but it is nevertheless used to determine the lowest parallel configuration required.

Each Samsung 50E cell has a maximum CDR of 9.8A. In addition, the powercell also has a battery capacity of 5Ah. As explained previously, adding two or more cells in parallel increases both the

amount of current the battery is capable of delivering and its cell-group capacity, however voltage remains the same.

The required current is 20A max continuous. Therefore three cells were added in parallel to give a max CDR of 29.4A and capacity of 15Ah. The calculations are shown below.

$$\begin{aligned}
&\text{1 cell parallel group:} && 1 \times 9.8\,\text{A} = 9.8\,\text{A (max CDR)}, && 1 \times 5\,\text{Ah} = 5\,\text{Ah} \\
&\text{2 cell parallel group:} && 2 \times 9.8\,\text{A} = 19.6\,\text{A (max CDR)}, && 2 \times 5\,\text{Ah} = 10\,\text{Ah} \\
&\text{3 cell parallel group:} && 3 \times 9.8\,\text{A} = 29.4\,\text{A (max CDR)}, && 3 \times 5\,\text{Ah} = 15\,\text{Ah}
\end{aligned}$$

Figure 3: Current and capacity values for 1, 2, and 3 cell parallel groups.

A two cell parallel group would only be capable of delivering 19.6A, which falls short of the desired 20A. Hence, a 7S3P battery configuration is chosen. From Equation 3, total energy E of the pack is computed as follows.

$$\begin{aligned}
E &= 24\,\text{V} \times 15\,\text{Ah} \\
E &= 360\,\text{Wh}
\end{aligned}$$

Figure 4: Calculation of total energy capacity for the battery pack.

Furthermore, the BMS current rating is calculated in order to determine the appropriate BMS for this application. The max current the load will draw is 20A. Therefore, the BMS must be able to allow 20A to pass through it. An additional 15% will be added to the max current draw to give a little bit of headroom [1]. As a result, the BMS rating was calculated to be 23A. A summary of the results is presented in the table below.

Parameter	**Value**
Pack configuration	7S3P
BMS rating	23A
Total Pack energy	360Wh
Nominal pack voltage	24V
Pack capacity	15Ah

Table 4: Summary of Results for the Battery Pack Configuration

CONCLUSIONS

The chosen e-bike battery pack for this project is a 7S3P configuration, with a total energy of 360Wh. Typically, general e-bike battery packs range between 300Wh to 1000Wh [3]. Battery voltages for e-bikes fall under the 24V, 36V, 48V, 50V or 52V levels with various capacities. Thus, this battery pack is on the lower side. Furthermore, e-bikes can be categorised in various forms such as battery type, mode of propulsion, motor type, by application, among others [4]. In terms of mode of propulsion, e-bikes can either be pedal-assist or fully throttle. As the name suggests, pedal assist e-bikes boost the rider's pedaling while throttle e-bikes run solely by the electric motor [4]. As a result of this, pedal assist e-bikes consume less energy as the rider also puts in some work. The chosen mode of propulsion for this e-bike project will be pedal-assist, as it uses less energy, permits the user to still experience the thrill of riding a bicycle, and is cost-effective.

Equally important, it may be important to discuss the estimated range from the stated 360Wh battery. On average, most ebike motors consume about 10-20 Wh/mile with moderate assistance [5]. Thus, a 360Wh pack should be able to give 18-36 miles of range on a single charge. For comparison, a Jaison EB5 e-bike sold on Amazon with pedal-assist sensor (PAS), 500W motor, and a battery capacity of 360Wh covers a range of up to 25-40 miles per charge. One thing to be aware of is that e-bikes in the market typically come with both PAS and throttle mode, where the lower range value indicates when the bicycle is being used in a high-assist or purely electric mode. The Jaison EB5 has a 36V battery with 10Ah of capacity, which means it has a 10S configuration. The number of parallel cells cannot be determined since the type of cells used are not known. The Jaison EB5 has a max speed of 20mph and costs about $400 pre-tax.

To put things into perspective, the Tesla Model 3 rear-wheel drive (RWD) has a battery capacity of 57.5kWh [6], which would be about 160 times the battery capacity of this e-bike project. Considering that the car weighs 1760 kg [7] with no passengers, a tremendous amount of energy is needed to propel it. E-bikes on the other hand weigh 26-32 kg on average [8], which means that the rider adds significantly more weight. The Tesla Model 3 RWD has an estimated 308 miles (495 km) of

range [6].

In summary, as the need to reduce greenhouse gas emissions becomes more urgent, consumers are looking towards lightweight, affordable, and eco-friendly alternatives such as e-bikes in order to reduce their carbon footprint. Understanding the primary energy source of electric vehicles is crucial to ensuring a successful transition, where preservation and sustainability will remain guiding pillars.

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[5] LiamNina. "How far can an electric bike go on one charge?" (2025), [Online]. Available: https://eu.fiido.com/blogs/news/how-far-can-an-electric-bike-go-on-one-charge#:~:text=But%2C%20on%20average%20terrain%2C%20most,%2Fmile)%20and%20moderate%20assistance. (visited on 06/09/2025).

[6] EVBox. "How long does it take to charge the tesla model 3?" (), [Online]. Available: https://evbox.com/uk-en/electric-cars/tesla/tesla-model-3#:~:text=The%20Model%203%20Rear%2DWheel,maximum%20power%20of%2011%20kW. (visited on 06/09/2025).

[7] Tesla. "Model 3 owner's manual | tesla." (2024), [Online]. Available: https://www.tesla.com/ownersmanual/model3/en_cn/GUID-56562137-FC31-4110-A13C-9A9FC6657BF0.html (visited on 06/09/2025).

[8] Gazellebikes. "Weight of an ebike." (), [Online]. Available: https://www.gazellebikes.com/en-us/knowledge-base/electric-bikes/weight#:~:text=An%20electric%20bike%20without%20the,heavier%20than%20a%20normal%20bike. (visited on 06/09/2025).

APPENDIX

The following pages show the INR21700-50E data sheet.

Specification of Product

Tentative

1. Customer :

2. Product : Lithium-ion Rechargeable Cell

3. SDI Model : INR21700-50E

4. Approved by

Division			
Signature			
Date	/ /	/ /	/ /

5. Date of Application (YY/MM/DD) : 2017/12/19

6. Supplier : **SAMSUNG SDI Co., Ltd.**
Battery Business Division

Issued	Checked	Approved

Revision history

Revision No.	Date ('yr-m-d)	Page	Item	Description	Changes / Author
v0.0	'17-10-13			First version (Tentative)	JH, Lee
v0.1	'17-10-31			Packaging, condition update	JH, Lee Lucy Kim
v0.2	'17-12-07			Add Standard characteristics (Tentative ver.)	SoonChul Byun Wayne Kim

1. Scope

This product specification has been prepared to specify the rechargeable lithium-ion cell ('cell') to be supplied to the customer by Samsung SDI Co., Ltd.

2. Description and Model

2.1 Description Cell (lithium-ion rechargeable cell)
2.2 Model INR21700-50E
2.3 Site Manufactured in Korea

3. Nominal Specifications (*1)

Item	Specification
3.1 Standard discharge Capacity	Min 4,900mAh - Charge : 0.5C(2,450mA), 4.2V, 0.02C(98mA) cutoff @ RT - Discharge : 0.2C(980mA), 2.5V cutoff @ RT * 1C = 4,900mA
3.2 Rated discharge Capacity	Min 4,753mAh - Charge : 0.5C(2,450mA), 4.2V, 0.02C(98mA) cutoff @ RT - Discharge : 1C(4,900mA), 2.5V cutoff @ RT
3.3 Charging Voltage	4.2V
3.4 Nominal Voltage	3.6V
3.5 Charging Method	CC-CV (constant voltage with limited current)
3.6 Charging Current	Standard charge: 2,450mA
3.7 Charging Time	Standard charge: 3hours
3.8 Max. Charge Current	4,900mA (not for cycle life)
3.9 Max. Discharge Current	9,800mA (for continuous discharge) 14,700mA (not for continuous discharge)
3.10 Discharge Cut-off Voltage	2.5V
3.11 Cycle life	Capacity ≥ 3,802mAh @ after 500cycles (80% of the Rated Discharge Capacity @ RT) - Charge : 0.5C(2,450mA), 4.2V, CCCV 0.05C(245mA) cut-off @ RT - Discharge: 1C(4,900mA), 2.5V cut-off @ RT
3.13 Recovery characteristics	Capacity recovery (after the storage) ≥ 4,278mAh (90% of the Rated Discharge Capacity @ RT) - Charge : 0.5C(2,450mA), 4.2V, 0.02C(98mA) cutoff @ RT - Storage : 30 days @ 60℃ - Discharge : 1.0C(4,900mA) 2.50V cut-off @ RT
3.14 Cell Weight	69g max
3.15 Cell Dimension	Cell height : Max.70.80mm Diameter : Φ Max.20.25mm

3.16 Operating Temperature(*2) (Cell Surface Temperature)	Charge : 0 to 45°C Discharge : -20 to 60°C
3.17 Storage Temperature (*3)	1 year : -20~23°C 3 months : -20~45°C 1 month : -20~60°C

Note (*1): Protection function of the battery pack should be set within the specified charge, discharge and temperature range in the Cell Specification.

Note (*2): Discharge OTP(over temp. protection) should not be over 60'C of the cell surface temperature. Protection set should be based on the location of the cell surface with the highest temp increase part of the battery pack

Note (*3): If the cell is kept as ex-factory status (30% of charge), the capacity recovery rate is more than 80%.

4. Outline Dimensions

See the Fig. 1

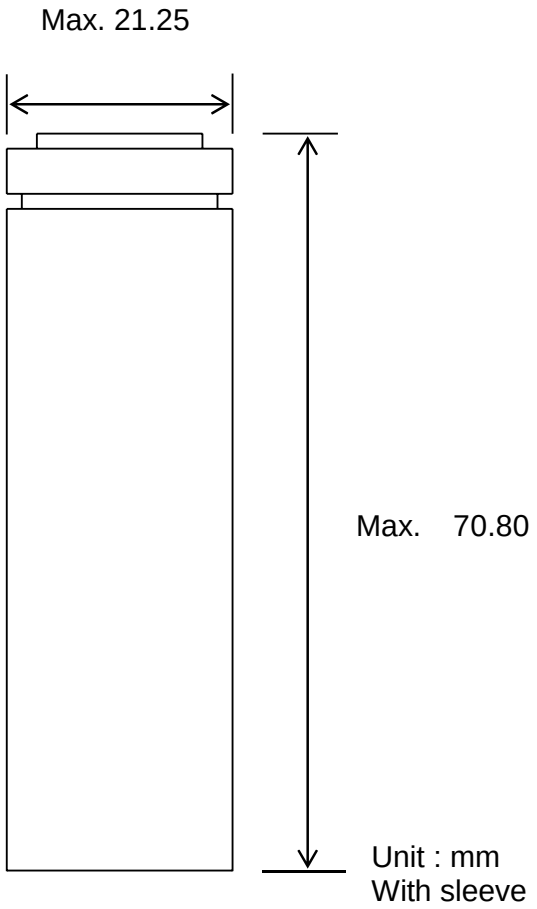

Fig. 1 Outline Dimensions of INR21700-50E

5. Appearance

There shall be no such defects as scratch, rust, discoloration, leakage which may adversely affect commercial value of the cell.

6. Standard Test Conditions

6.1 Environmental Conditions

Unless otherwise specified, all tests stated in this specification are conducted at temperature 23±3°(@RT)and humidity under 65%.

6.2 Measuring Equipment

(1) Amp-meter and Volt-meter

The amp-meter and volt-meter should have an accuracy of the grade 0.5mA and mV or higher.

(2) Slide caliper

The slide caliper should have 0.01 mm scale.

(3) Impedance meter

The impedance meter with AC 1kHz should be used.

7. Characteristics

7.1 Standard Charge

This "Standard Charge" means charging the cell with charge current of 0.5C (2,450mA) and constant voltage 4.2V and 0.02C(98mA) cut-off in CV mode at 23°C.

7.2 Standard Discharge Capacity

The standard discharge capacity is the initial discharge capacity of the cell, which is measured with discharge current of 0.2C(980mA) with 2.50V cut-off at 23°C within 1 hour after the Standard charge.

Standard Discharge Capacity ≥ 4,900mAh

7.3 Rated Discharge Capacity

The rated discharge capacity is the initial discharge capacity of the cell, which is measured with discharge current of 1C(4,900mA) with 2.50V cut-off at 23°C within 1 hour after the Standard charge.

Rated Discharge Capacity ≥ 4,753mAh (97% of 4,900mAh)

7.4 Initial internal impedance

Initial internal impedance measured at AC 1kHz after Standard charge.

Initial internal impedance ≤ 35mΩ

7.5 Temperature Dependence of Discharge Capacity

Discharge capacity comparison at each temperature, measured with discharge constant current 1C(4,900mA) and 2.50V cut-off with follow temperature after the standard charging at 23°C.

Charge Temperature	Discharge temperature		
23°C	-10°C	23°C	40°C
Relative Capacity	60%	97%	97%

Note: If charge temperature and discharge temperature is not the same, the interval for temperature change is 3 hours.
Percentage as an index of the Standard discharge capacity (=4,900mAh) at 23°C is 100%.

7.6 Discharge Rate Capabilities

Discharge capacity is measured with the various currents in under table and 2.50V cut-off after the Standard charge at 23°C.

	Discharge condition		
Current	0.2C (980mA)	1C (4900mA)	2C (9800mA)
Relative Capacity	100%	97%	95%

Note: Percentage as an index of the Standard discharge capacity (=4,900mAh) is 100%.

7.7 Cycle Life

Each cycle is an interval between the charge (charge current 2,450mA) with 0.05C(245mA) cut-off and the discharge (discharge current 4,900mA) with 2.50V cut-off. Capacity after 500cycles.

Capacity ≥ 3,802mAh (80% of Rated Capacity)

7.9 Recovery Characteristics

Capacity after storage for 30days at 60°C after the Standard charged at 23°C, measured with discharge current 4,900mA with 2.50V cut-off at 23℃.

Capacity recovery(after the storage) ≥ 4,278mAh (90% of Rated Capacity)

7.10 Status of the cell as of ex-factory

The cell should be shipped in 3.43V ~ 3.63V Charging voltage range.

8. Mechanical Characteristics

8.1 Drop Test

Test method: Each fully charged cell or battery is dropped three times from a height of 1.0 m onto a concrete floor. The cells or batteries are dropped so as to obtain impacts in random orientations. After the test, the sample shall be put on rest for a minimum of one hour and then a visual inspection shall be performed.

Criteria: No fire, no explosion

Drop test shall be performed with the IEC 62133

8.2 Vibration Test

Test method: As to the UN transportation regulation(UN38.3), for each axis (X and Y axis with cylindrical cells) 7Hz→200Hz→7Hz for 15min, repetition 12 times totally 3hours, the acceleration 1g during 7 to 18Hz and 8g (amplitude 1.6mm) up to 200Hz.

Criteria: No leakage, with less than 10mV of OCV drop

Vibration test shall be performed with the UN38.3 standard

9. Safety

9.1 Overcharge test

Test method: To charge the standard charged cell with 4.2V and 3C (7,35A) at 23°C for 7 hours

Criteria: No fire, and no explosion.

Overcharge test shall be performed with the UL1642 standard

9.2 External short-circuit test

Test method: Fully rated charged cell is to be short-circuited by connecting the positive and negative terminals of the battery with a circuit load having a resistance load of 80 ±20 mΩ. The battery is to discharge until a fire or explosion is obtained, or until it has reached a completely discharged state of less than 0.2 V and the battery case temperature has returned to ±10°C of ambient temperature. The return to near ambient of the battery (cell) casing in an indication of ultimate results.
Tests are to be conducted at 20 ±5°C and at 55 ±5°C.

Criteria: No fire, and no explosion.

External short-circuit test shall be performed with the UL1642 standard

9.3 Forced discharge test

Test method: A discharged cell is subjected to a reverse charge at 1.0C(4,900mA) for 90 min.

Criteria: No fire, and no explosion.

Forced discharge test shall be performed with the IEC62133 standard

9.4 Heating test

Test method: To heat up the standard charged cell at heating rate 5℃ per minute up to 130℃ and keep the cell in oven for 30 min.

Criteria: No fire, and no explosion.
Heating test shall be performed with the UL1642 standard(10 min) & GB31241 standard(30 min).

10. Warranty

Samsung SDI will be responsible for replacing the cell against defects or poor workmanship for 15 months from the date of shipping. Any other problems caused by malfunction of the equipment or unsuitable use of the cell are not under this warranty.
The warranty set forth in proper use, handling conditions described above,
and excludes in the case of a defect which is not related to manufacturing of the cell.

11. Others

11.1 Storage for a long time

If the cell is kept for a long time (3months or more), It is strongly recommended that the cell is preserved at dry and low-temperature.

11.2 Other

Any matters that specifications does not have, should be conferred with between the both parties.

12. Packaging

See Fig.2: Package Drawing

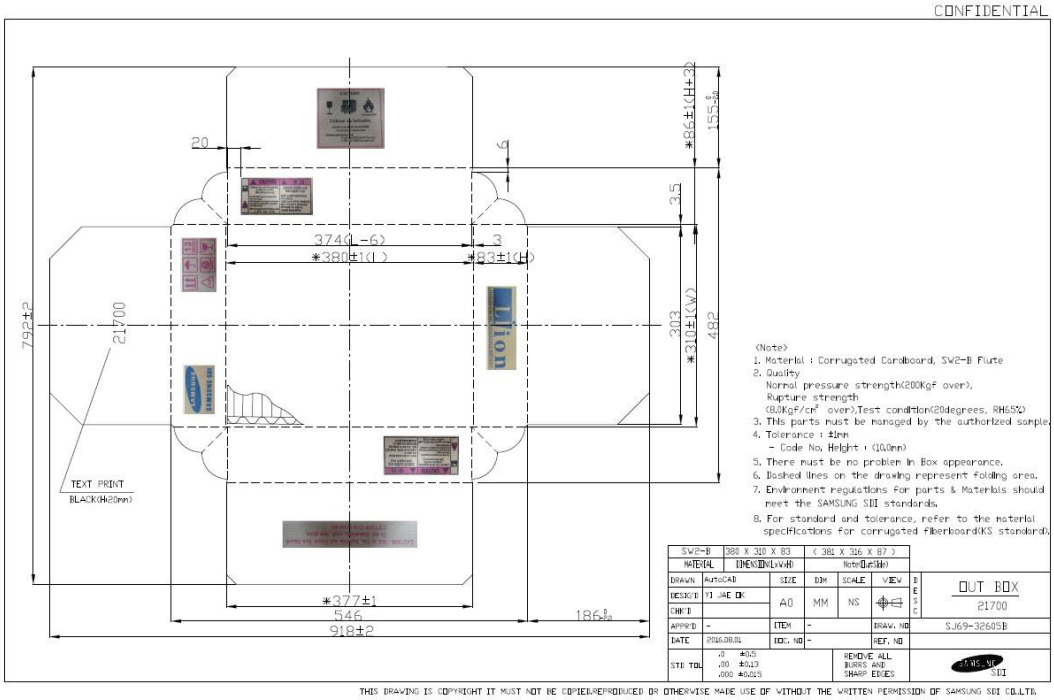

Fig. 2-1. Outer box package drawing.

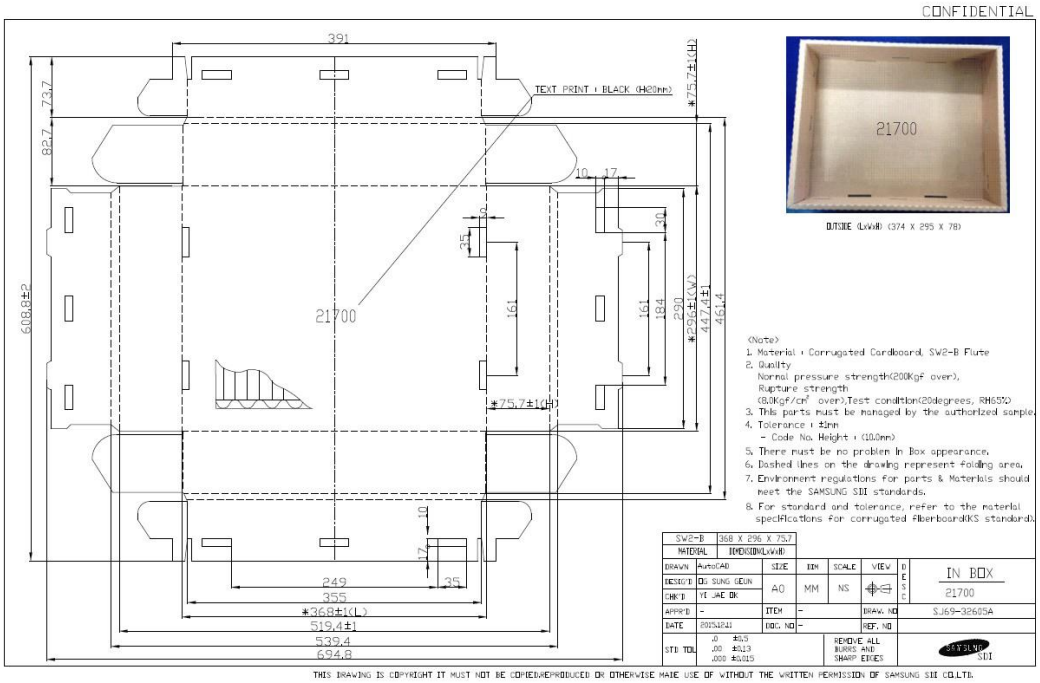

Fig. 2-2. Inner box package drawing.

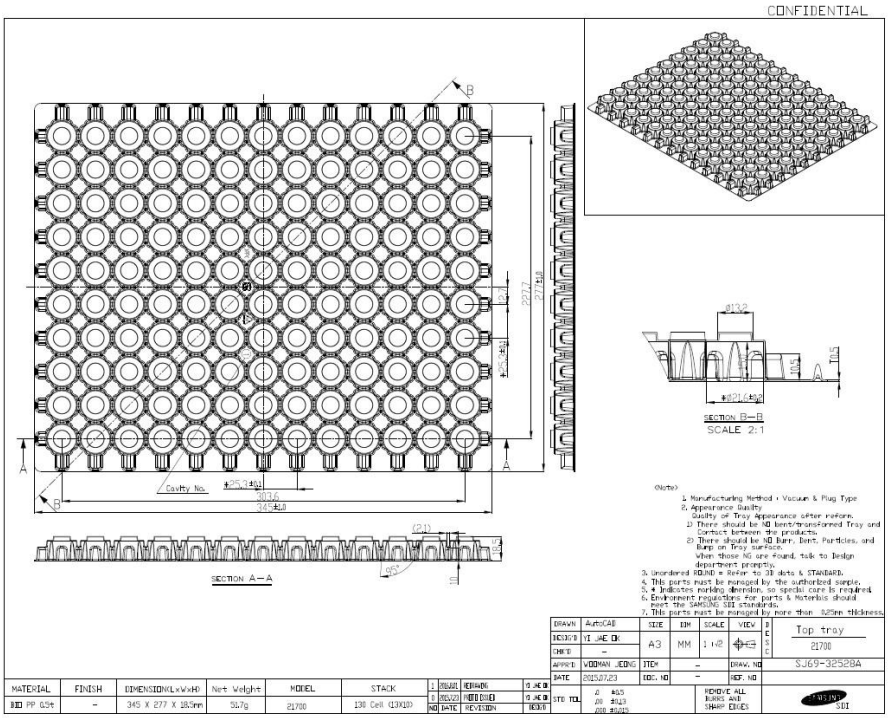

Fig. 2-3. Upper tray drawing.

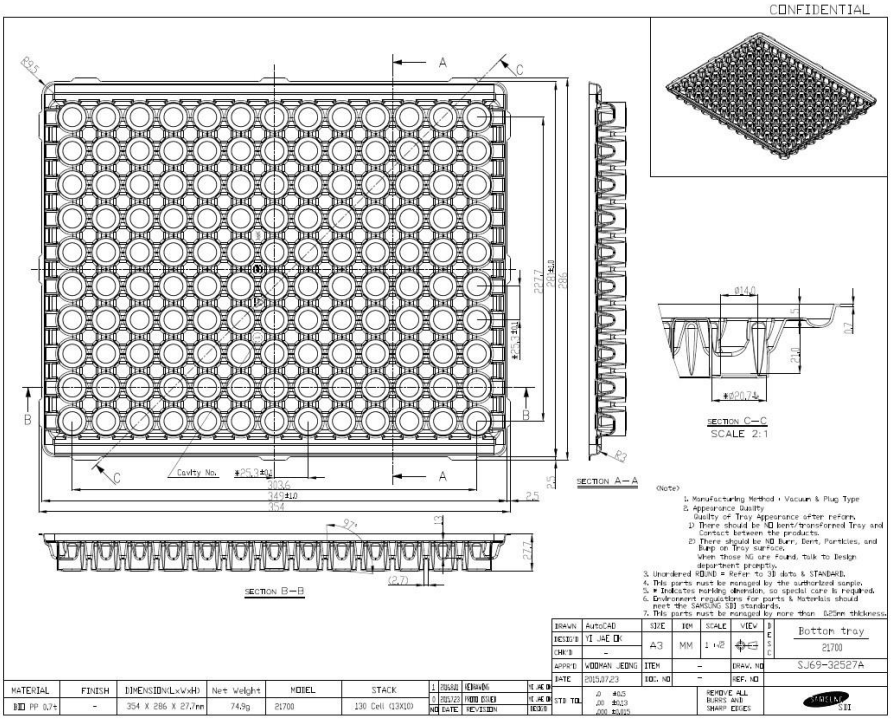

Fig. 2-4. Bottom tray drawing.

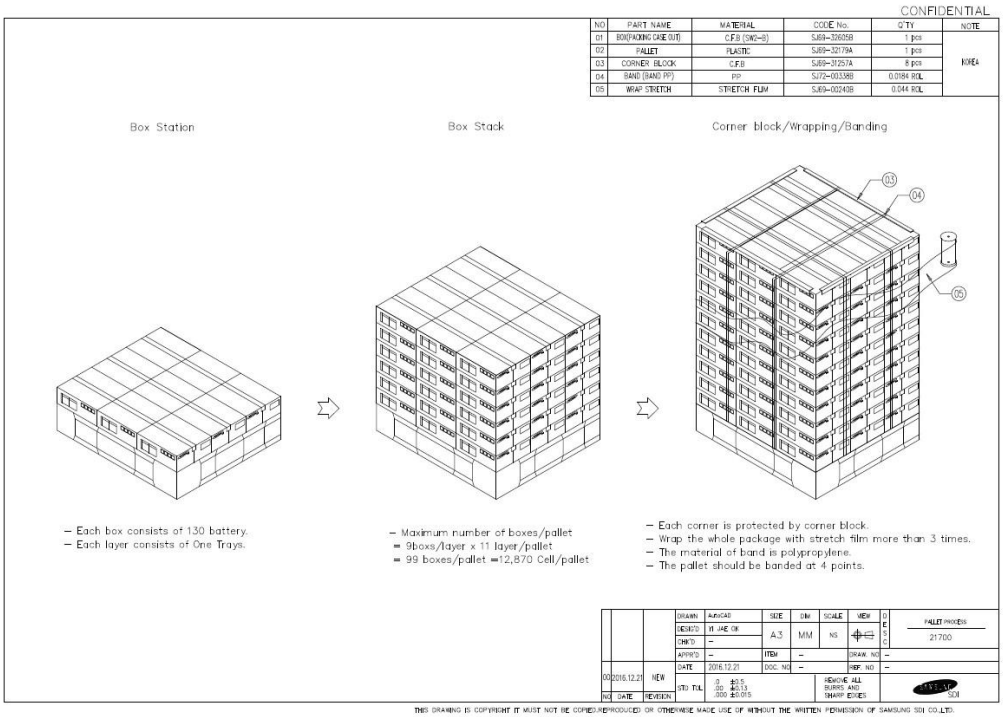

Fig. 2-5. Pallet packaging process

Proper Use and Handling of Lithium Ion Cells

See before using lithium-ion cell

This document has been prepared to describe the appropriate cautions and prohibitions, which the customer should take or employ when the customer uses and handles the lithium ion cell to be manufactured and supplied by Samsung SDI Co., Ltd., in order to obtain optimum performance and safety.

1. Charging

1.1 Charging current

Charging current should be less than maximum charge current specified in the product specification.

1.2 Charging voltage

Charging should be done by voltage less than that specified in the product specification.

1.3 Charging time

Continuous charging under specified voltage does not cause any loss of performance characteristics. However, the charge timer is recommended to be installed from a safety consideration, which shuts off further charging at time specified in the product specification.

1.4 Charging temperature

The cell should be charged within a range of specified temperatures in the specification.

1.5 Reverse charging

The cell should be connected, confirming that its poles are correctly aligned. Inverse charging should be strictly prohibited. If the cell is connected improperly, it may be damaged.

2. Discharging

2.1 Discharging

The cell shall be discharged continuously at less than maximum discharge current specified in the product specification. In case of the higher discharge current should be set, it shall be discussed together with SDI.

2.2 Discharging temperature

2.2.1 The cell should be discharged within a range of temperatures specified in the product specification.

2.2.2 Otherwise, it may cause loss of characteristics.

2.3 Over-discharging

2.3.1 The system should be equipped with a device to prevent further discharging exceeding discharging cut-off voltage specified in the product specification.

2.3.2 Over-discharging may cause loss of performance, characteristics of battery function.

2.3.3 Over-discharging may occur by self-discharge if the battery is left for a very long time without any use.

2.3.4 The charger should be equipped with a device to detect cell voltage and to determine recharging procedures.

3. Storage

3.1 Storage conditions

3.1.1 The cell should be stored within a range of temperatures specified in the product specification.

3.1.2 Otherwise, it may cause loss of performance characteristics, leakage and/or rust.

3.2 Long-term storage

3.2.1 The cell should be used within a short period after charging because long-term storage may cause loss of capacity by self-discharging.

3.2.2 If long-term storage is necessary, the cell should be stored at lower voltage within a range specified in the product specification, because storage with higher voltage may cause more loss of performance characteristics.

3.2.3 Store the battery at 0 ~ 23℃, low humidity (below 65%), no dust and no corrosive gas atmosphere.

4. Cycle life

4.1 Cycle life performance

4.1.1 The cell can be charged/discharged repeatedly up to times specified in the produce specification with a certain level of capacity also specified in the product specification.

4.1.2 Cycle life may be determined by conditions of charging, discharging, operating temperature and/or storage.

5. Battery Pack Assembly

5.1 Prohibition of usage of damaged cell

5.1.1 The cell should be inspected visually before battery assembly.

5.1.2 The cell should not be used if sleeve-damage, can-distortion and/or electrolyte-smell is detected.

5.2 Terminals handling

5.2.1 Excessive force on the negative terminal should be avoided when external strip terminal is welled.

5.3 Transportation

5.3.1 If the cell is necessary to be transported to such as the battery manufacturer, careful precautions should be taken to avoid damage of cell.

6. Others

6.1 Disassembly

6.1.1 The cell should not be dismantled from the battery pack.

6.1.2 Internal short-circuit caused by disassembly may lead to heat generation and/or venting.

6.1.3 When the electrolyte is coming in contact with the skin or eyes, wash immediately with fresh water and seek medical advice.

6.2 Short-circuiting

6.2.1 Short-circuit results in very high current which leads to heat generation.

6.2.2 An appropriate circuitry should be employed to protect accidental short-circuiting.

6.3 Incineration

6.3.1 Incinerating and disposing of the cell in fire are strictly prohibited, because it may cause rupture and explosion.

6.4 Immersion

6.4.1 Soaking the cell in water is strictly prohibited, because it may cause corrosion and leakage of components to be damaged to functions.

6.5 Mixing use

6.5.1 Different types of cell, or same types but different cell manufacturer's shall not be used, which may lead to cell imbalance, cell rupture or damage to system due to the different characteristics of cell.

6.5.2 Do not mix use the cells from different batches/ranks even they are the same cell types from the same manufacturer.

6.6 Battery disposal

6.6.1 Although the cell contains no environmentally hazardous component, such as lead or cadmium. the battery should be disposed according to the local regulations when it is disposed.

6.6.2 The cell should be disposed with a discharged state to avoid heat generation by an inadvertent short-circuit.

6.7 Caution

6.7.1 The Battery used in this device may present a risk of fire or chemical burn if mistreated.

6.7.2 Do not disassemble, expose to heat above 100℃ or incinerate it.

6.7.3 Replace battery with Samsung SDI battery only.

6.7.4 Use of another battery may present a risk of fire or explosion.

6.7.5 Dispose of used battery promptly.

6.7.6 Keep away from children.

6.7.7 Do not disassemble and do not dispose of in fire.

Pack Design Guideline for Lithium Ion Cells

If a customer's battery pack cannot meet Pack Design Guideline,
SDI cannot take responsibility for a quality issue about a battery cell.

1. Electrical design

1.1 Charge

1.1.1 Stop charging the battery if charging isn't completed within the specified time.
1.1.2 The battery can be used within the following temperature ranges.
Don't exceed these ranges.
Charge temperature ranges : 0℃ ~ 45℃
1.1.3 Charge current must be controlled by specified value in Cell specification.
1.1.4 Cut-off Voltage of charging must be under 4.2V.
1.1.5 Charger must stop charging battery by detecting either charging time or current specified in Cell's specification.
1.1.6 Recharge Voltage must be at least 0.1V lower than Max charging Voltage.
1.1.7 Do not charge the battery under 1.0V voltage.
1.1.8 Voltage range of the battery for pre-charge mode is 1.0 ~ 3.0V with a charging current of 0.1 ~ 0.5C

1.2 Discharge

1.2.1 Discharge current must be controlled by specified value in Cell's specification.
1.2.2 Cut-off Voltage of discharging must be over 2.5V.
1.2.3 The battery can be used within the following temperature ranges.
Don't exceed these ranges.
Discharge temperature ranges : -20℃ ~ 60℃

1.3 Storage

Store the battery at low temperature (below 60℃, 20℃ is recommended), low humidity(below 65%), no dust and no corrosive gas atmosphere.

1.4 Design of positioning the battery pack in application and charger

To prevent the deterioration of the battery performance caused by heat, battery shall be positioned away from the area where heat is generated in the application and the charger.

1.5 Safety Device

Be sure adopting proper safe device such as PTC specified type or model in Cell Specification. If you intend to adopt different safety device which is not specified in Cell Specification, please contact Samsung SDI to investigate any potential safety problem. Be sure designing 2nd protective devices such as PTC & PCM at the same time to protect Cell just in case one protective device is fault.

2. Design of System

2.1 Connection between the cell and the battery

2.1.1 The cell should not be soldered directly with other cells. Namely, the cell should be welded with leads on its terminal and then be soldered with wire or leads to solder.

2.1.2 Otherwise, it may cause damage of component, such as separator and insulator, by heat generation.

2.2 Positioning the battery in the System

2.2.1 The battery should be positioned as possible as far from heat sources and high temperature components.

2.2.2 Otherwise, it may cause loss of characteristics.

2.2.3 The recommended spacing between the cells is more than 1mm.

2.3 Mechanical shock protection of the battery

2.3.1 The battery should be equipped with appropriate shock absorbers in the pack in order to minimize shock, which can damage the cells. .

2.3.2 Otherwise, it may cause shape distortion, leakage, heat generation and/or rupture and/or open circuit.

2.3.3 Ultrasonic should not be used when manufacturing battery packs as it can damage the connection between component parts.

2.4 Short-circuit protection of the cell

2.4.1 The cell is equipped with an insulating sleeve to protect short-circuit which may occur during transportation, battery assembly and /or system operation.

2.4.2 If the cell sleeve is damaged by some causes such as outside impact, it may cause short-circuit with some wiring inside the battery.

2.5 Connection between the battery and charger/system

2.5.1 The battery should be designed to be connected only to the specified charger and system.

2.5.2 A reverse connection of the battery, even in the specified system, should be avoided by employing special battery design such as a special terminals.

2.6 Pack design

2.6.1 The current consumption of the battery pack should be under 10uA after shut down.

2.6.2 Cell voltage monitoring system.
The system (charger or pack) should be equipped with a device to monitor each voltage of cell block to avoid cell imbalance which can cause damage to the cells.

2.6.3 The battery pack or system should have warning system such as over temperature, over voltage, over current, and so on.

2.6.4 Mechanical design

Item	Portable IT	Power-Tool	Medical	E-Bike	E-Scooter	EV/LEV	ESS/UPS
☐ Need partitions(separator) between BMS and a Cell	◎	◎	◎	◎	◎	◎	◎
☐ Need to check if the pack is designed to be able to avoid thermal runaway (1KWh ↑)					◎	◎	◎
☐ Need to analyze the battery pack's thermal distribution and its effect on the pack's life cycle					◎	◎	◎
☐ Need to use a non-flammable (VO level) case	◎	◎	◎	◎	◎	◎	◎
☐ Need to apply improved material (Steel) to the case						◎	◎
☐ Need to analyze the battery pack's structure, system, installation status and use environment						◎	◎

Handling precaution and prohibitions of lithium rechargeable cells and batteries

Inaccurate handling of lithium ion and lithium ion batteries rechargeable battery may cause leakage, heat, smoke, an explosion, or fire.
This could cause deterioration of performance or failure. Please be sure to follow instructions carefully.

1. Safety precaution and prohibitions

To assure product safety, describe the following precautions in the instruction manual of the application.

2. Danger

2.1 Electrical misusage

2.1.1 Use dedicated charger.
2.1.2 Use or charge the battery only in the dedicated application.
2.1.3 Don't charge the battery by an electric outlet directly or a cigarette lighter charger.
2.1.4 Don't charge the battery reversely.

2.2 Environmental misusage

2.2.1 Don't leave the battery near the fire or a heated source.
2.2.2 Don't throw the battery into the fire.
2.2.3 Don't leave, charge or use the battery in a car or similar place where inside of temperature may be over 60℃.
2.2.4 Don't immerse, throw, wet the battery in water / seawater.

2.3 Others

2.3.1 Don't fold the battery cased with laminated film such as pouch and Polymer.
2.3.2 Don't store the battery in a pocket or a bag together with metallic objects such as keys, necklaces, hairpins, coins, or screws.
2.3.3 Don't short circuit (+) and (-) terminals with metallic object intentionally.
2.3.4 Don't pierce the battery with a sharp object such as a needle, screw drivers.
2.3.5 Don't heat partial area of the battery with heated objects such as soldering iron.
2.3.6 Don't hit with heavy objects such as a hammer, weight.
2.3.7 Don't step on the battery and throw or drop the battery on the hard floor to avoid mechanical hock.
2.3.8 Don't disassemble the battery or modify the battery design including electric circuit.
2.3.9 Don't solder on the battery directly.
2.3.10 Don't use seriously scared or deformed battery.
2.3.11 Don't put the battery into a microwave oven, dryer ,or high-pressure container.
2.3.12 Don't use or assemble the battery with other makers' batteries, different types and/or models of batteries such as dry batteries, nickel-metal hydride batteries, or nickel-cadmium batteries.
2.3.13 Don't use or assemble old and new batteries together.

3. Warning

3.1 Stop using the battery if the battery becomes abnormally hot, order, discoloration, deformation, or abnormal conditions is detected during use, charge, or storage.

3.2 Keep away from fire immediately when leakage or foul odors are detected. If liquid leaks onto your skin or cloths, wash well with fresh water immediately.

3.3 If liquid leaking from the battery gets into your eyes, don't rub your eyes and wash them with clean water and go to see a doctor immediately.

3.4 If the terminals of the battery become dirty, wipe with a dry cloth before using the battery.

3.5 Cover terminals with proper insulating tape before disposal.

4. Caution

4.1 Electrical misusage

Battery must be charge with constant current-constant voltage (CC/CV).

4.2 Others

4.2.1 Keep the battery away from babies and children to avoid any accidents such as swallow.

4.2.2 If younger children use the battery, their guardians should explain the proper handling method and precaution before using.

4.2.3 Before using the battery, be sure to read the user's manual and precaution of its handling.

4.2.4 Before using charger, be sure to read the user's manual of the charger.

4.2.5 Before installing and removing the battery from application, be sure to read user's manual of the application.

4.2.6 Replace the battery when using time of battery becomes much shorter than usual.

4.2.7 Cover terminals with insulating tape before proper disposal.

4.2.8 If the battery is needed to be stored for an long period, battery should be removed from the application and stored in a place where humidity and temperature are low.

4.2.9 While the battery is charged, used and stored, keep it away from object materials with static electric chargers.

5. Safety Handling Procedure for the Transporter

5.1 Quarantine

Packages that are crushed, punctured or torn open to reveal contents should not be transported. Such packages should be isolated until the shipper has been consulted, provided instructions and, if appropriate, arranged to have the product inspected and repacked.

5.2 Spilled Product

In the event that damage to packaging results in the release of cells or batteries, the spilled products should be promptly collected and segregated and the shipper should be contacted for instructions.

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